RAPID Project Submission

Phishing Detection Model

This project trains a Random Forest classifier to detect phishing activity from the refined dataset. The final model was generated successfully and saved in the project directory.

Project Overview

• Dataset: refined_phishing_dataset.xlsx
• Model: RandomForestClassifier
• Target: status (phishing vs legitimate)
• Output artifact: best_phishing_model.pkl

Model Performance

Test Accuracy: 96.28%

The model was trained using an 80/20 train-test split with stratification and achieved a strong phishing detection score.

Training Script

```
import joblib
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split

df_refined = pd.read_excel('refined_phishing_dataset.xlsx')
X = df_refined.drop(columns=['status'])
y = df_refined['status'].apply(lambda x: 1 if x == 'phishing' else 0)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

model = RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
print(f'Model Training Complete! Test Accuracy: {acc * 100:.2f}%')

joblib.dump(model, 'best_phishing_model.pkl')
```

Files in Project Folder

• train_model.py
• refined_phishing_dataset.xlsx
• best_phishing_model.pkl
• RAPID_Project_Submission.pdf